NITHIN K

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SUMMARY

VLSI Design and Verification-focused ECE graduate from Johns Hopkins University with experience in FPGA, and ASIC design and functional verification. Skilled in Verilog and SystemVerilog languages with demonstrated work in AI hardware acceleration and in-memory computing. Seeking to contribute to the design and development of scalable, high-performance digital systems and next-generation computing solutions.

EDUCATION

JOHNS HOPKINS UNIVERSITY | Baltimore, MD
MSE in Electrical and Computer Engineering | Graduated - 05/2026 | 3.6/4.0

VELLORE INSTITUTE OF TECHNOLOGY, VIT | Tamil Nadu, India
B.Tech in Electrical and Electronics Engineering |Graduated - 07/2024 | 3.7/4.0

WORK EXPERIENCE

Digital Design Engineer | RND4IMPACT INC. | San Jose, CA (Remote)| June 2026 – Present

- Designing microarchitecture for a **RISC-V AI accelerator SoC** for Vision Mamba, defining the processor-accelerator interface, memory-mapped control registers, DMA dataflow, and execution control.
- Implementing parameterized **SystemVerilog RTL** modules including systolic GEMM engines, pipelined vector-processing datapaths, state machine control, and nonlinear function blocks.

Teaching Assistant: First-Year ECE Design | Johns Hopkins University | Baltimore, MD | January 2026 – May 2026

- Mentored 30+ students over weekly lab sections and office hours, teaching circuit fundamentals and electrical design concepts.
- Graded homework, quizzes, lab reports, and communicated technical feedback.

Digital Design Researcher – AI Hardware| Johns Hopkins University | Baltimore, MD | May 2025 - January 2026 | Project_Link

- Identified **latency bottlenecks** in the selective-scan computation of a Vision Mamba edge medical image classification model in **Python**.
- Designed **channel-parallel** 8-bit fixed-point **well-parameterized Verilog RTL** for the selective-scan kernel - including pipelined datapath, parallel MAC engine, memory, and state machines for control - reducing model latency by **9×** over the baseline.
- Implemented the accelerator on a **Kintex-7 FPGA**, meeting timing and resource utilization constraints in **AMD Vivado 2024.1**.
- **Published in IEEE Transactions on Biomedical Circuits and Systems**

NPU Verification Intern | STMicroelectronics | Uttar Pradesh, India | January 2024 - June 2024 | Project_Link

- Architected a complete 6T SRAM-based analog in-memory compute architecture in **Simulink**, auto-generating synthesizable **Verilog RTL** for functional verification using **Cadence Xcelium**.
- Developed LFSR-based noise injection models to emulate ±0.2% bit-cell noise and ±1 feature-level perturbations, stress-testing the architecture under **mixed-signal** operating conditions.

VLSI Design Intern | Maven Silicon Pvt. Ltd | Karnataka, India | May 2023 - July 2023 | Project_Link

- Implemented **Verilog RTL** for the AHB interface, APB controller **Finite-State Machine**, burst-transfer logic (INCR4/WRAP4), and Verilog testbench for an AMBA AHB-to-APB bridge in **Intel Quartus Prime**.

TECHNICAL PROJECTS

Investigating State-Space-based Vision Models for Memory-Efficient 3D Brain Tumor Segmentation| Johns Hopkins University | Baltimore, MD | January 2026 – May 2026 | Project_Link

- Replaced Transformer encoder blocks in **Swin-UNETR** with **Vision Mamba modules** to evaluate state-space models as memory-efficient alternatives to Transformers for 3D medical image segmentation, using **PyTorch/Python**.
- Achieved 20% reduction in **peak GPU training memory,** 46.6% reduction in **peak inference memory,** and **comparable Dice and Hausdorff Distance** scores relative to Swin-UNETR baseline.

Analog IC Design: 2-Bit 20 MS/s Flash ADC | Johns Hopkins University | Baltimore, MD | January 2025 - May 2025 | Project Link

- Enhanced a published 65 nm Flash ADC by redesigning for a 45 nm CMOS process on **Cadence Virtuoso**, introducing MOSCAP-based impedance matching and optimized sample-and-hold circuitry for improved rail-to-rail operation.

Design of 8-Bit & Binary Neural Network Accelerators| Johns Hopkins University | Baltimore, MD | August - December 2024 | Project Link

- Architected **RTL** of 8-bit and binarized neural network **ASIC accelerators**, comprising Fully Connected MAC engines, memory hierarchies, address-generation logic, nonlinear activations, and control datapaths for MNIST classification.
- Synthesized the designs using **Cadence Genus**, demonstrating an **~8% reduction** in combinational logic for the BNN architecture at equivalent area.
- Completed **place-and-route** in **Cadence Innovus** and analyzed post-layout PPA across combinational, sequential, memory, and I/O components.

CERTIFICATIONS

- **Cadence SystemVerilog for Design and Verification v25.03 Certification (Certified Feb 2026)**
  Integrated and verified a **VeriRISC CPU in SystemVerilog**, validating instruction execution, ALU functionality, memory operations, and program flow using directed diagnostic tests and a Fibonacci benchmark program.

- **Cadence SystemVerilog Assertions v5.2 Certification (Certified March 2026)**
  Architected **assertion-based verification for an SPI interface**, capturing protocol sequences, transaction headers, suspend handling, illegal transaction detection, and coverage-driven validation using SVA.

- **Cadence Essential SystemVerilog for UVM v1.2.5rev3 Certification (Certified June 2026)**
  Developed **Verification Components** to test a **4-port switch design**, covering **constrained-random** verification, virtual interfaces, sequencer, driver, and monitor subclasses through **Object-Oriented Programming** principles.

- **Cadence SystemVerilog Accelerated Verification with UVM v1.2.5rev2 (Certified July 2026)**
  Developed a **UVM test to verify a YAPP packet-router DUT**, incorporating configurable UVC agents with drivers, sequencers, and monitors; multichannel sequence generators, TLM scoreboarding, test, and sequence libraries.

- **Cadence Basic Static Timing Analysis v3.0 Certification (Expected July 2026)**
  Applying static timing analysis (STA) concepts by creating XDC timing constraints, analyzing setup/hold violations, and interpreting timing reports in AMD Vivado 2024.1.

- **Cadence Jasper Formal Fundamentals v25.09 Certification (Expected July 2026)**

- The Ultimate C++ Series Course **(Online Certification)**

RELEVANT COURSEWORK

- Mixed Mode VLSI Systems
- CAD Design of Digital VLSI Systems (Graduate Coursework)
- FPGA Synthesis Lab (Graduate Coursework)
- VLSI Design (Undergraduate Coursework)
- Computer Architecture and Organization (Undergraduate Coursework)
- Object-Oriented Programming with C++ (Undergraduate Coursework)
- Generative AI for Medical Imaging and Computer Vision (Graduate Coursework)
- Foundations of Reinforcement Learning (Graduate Coursework)